

Tree Borrows

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PLDI'25

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³MPI-SWS

Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {  
  
    *x = 13;  
  
    *y = 20;  
  
    *x  
  
}
```

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mutable thus disjoint

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mutable thus disjoint

*x has known value

*x is unchanged

always returns 13

Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {  
  
    *x = 13;  
  
    *y = 20;  
  
    13 // formerly *x: one fewer load from memory  
  
}
```

Type-level guarantees for references



`&mut` → mutation, no aliasing

`&` → aliasing, no mutation

Escape hatch: **unsafe**

Can use **unchecked operations** to do **low-level manipulations**

```
unsafe {  
    // Code within this block can effectively  
    // bypass some parts of the typechecker.  
    ...  
}
```

Within **unsafe** it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- ...

} violations trigger UB

Why have Undefined Behavior (UB) in a language?

Code that contains UB can have **any** behavior.

Relevance for optimizations:

allowed to change the behavior of programs that contain UB.

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Too little UB → weak optimizations

Too much UB → hard to write correct programs

What if **unsafe** code is misused ?

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    *x = 13;  
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}
```

```
fn main() {  
    let mut root = 42;  
    let ptr = &raw mut root;  
    let x = unsafe { &mut *ptr };  
    let y = unsafe { &mut *ptr };  
    println!("{}", write_both(x, y)); // prints 20  
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Not the compiler's responsibility

Within **unsafe** it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- compliance with aliasing rules ^{NEW!}

} violations trigger UB

Tree Borrows (TB): defines those aliasing rules

Not the compiler's responsibility

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- that pointers are non-null
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Sounds familiar?

Stacked Borrows has the same purpose,
Tree Borrows is its successor.

Stacked Borrows (SB)

In safe Rust, the Borrow Checker makes borrows well-bracketed. Stacked Borrows extends the well-bracketedness to **unsafe**.

```
let mut root = 42;  
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A diagram showing a memory stack. It consists of a single rectangular box with the word "root" inside. The box has a double-line border, suggesting it represents a memory location or pointer.

- new stack at root

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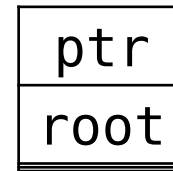
root

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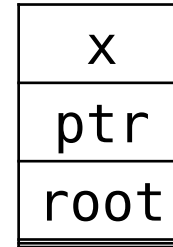


- ✓ root is at the top
- push ptr

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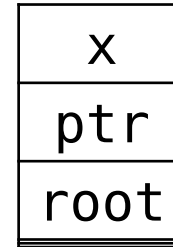


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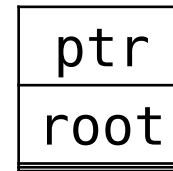


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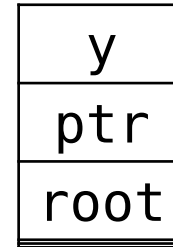


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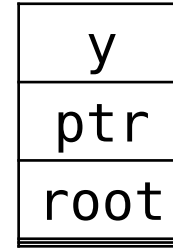


- pop until ptr is at the top
- push y

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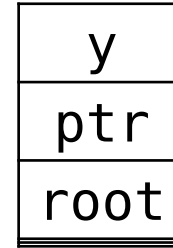
- search for x

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```

UB!



Can't use x if it is not in the stack

Desired outcome: UB

SB was **implemented** in Miri (official interpreter and UB detector)

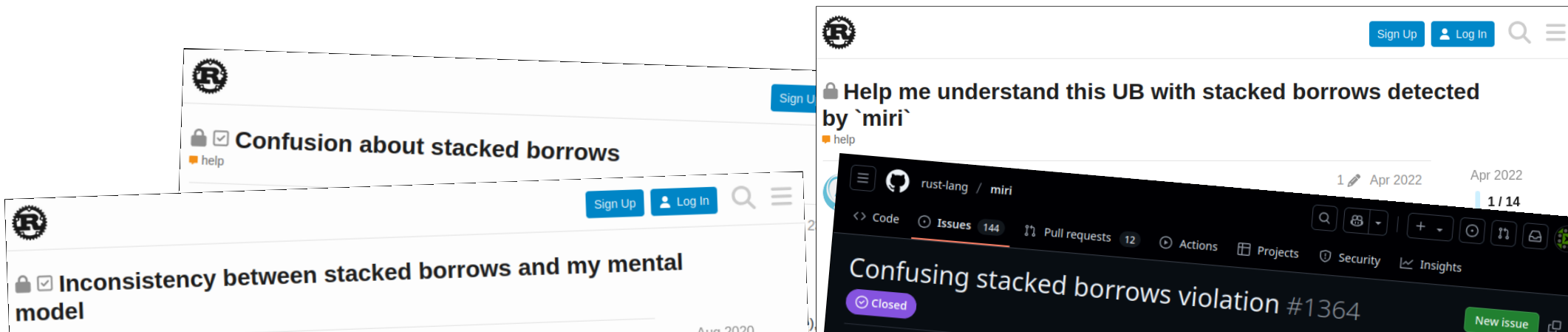
→ included in many projects' CI

→ several bugs detected (e.g. in stdlib)

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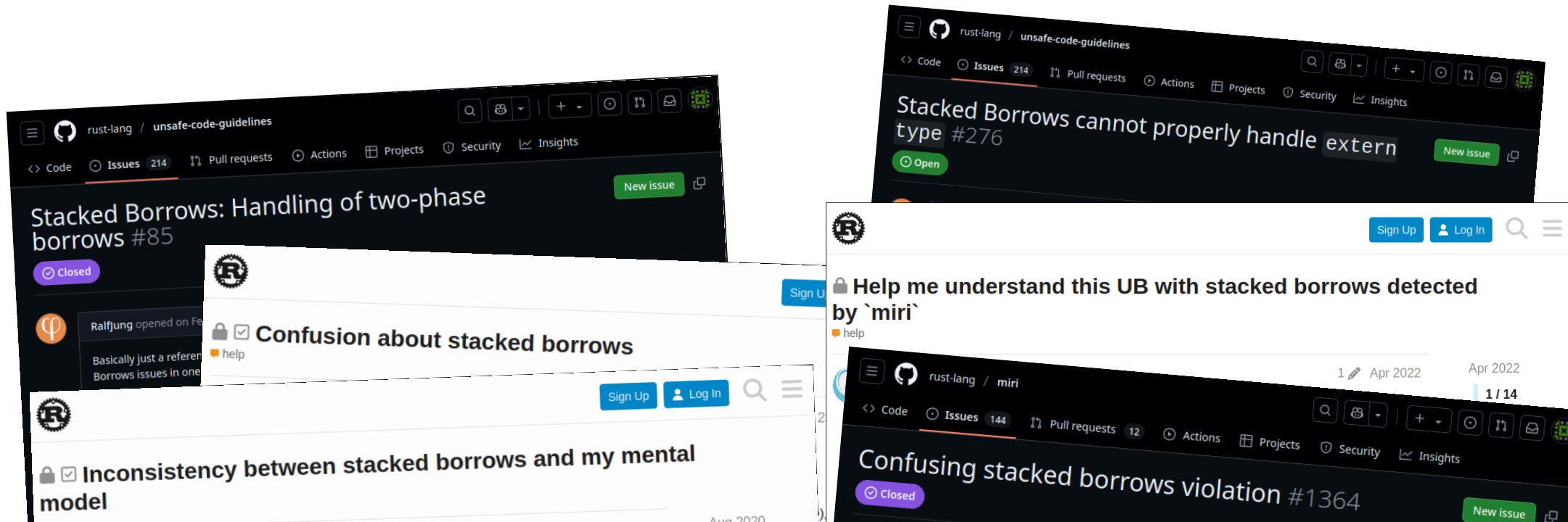
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→ included in many projects' CI

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Stacked Borrows: asserting uniqueness too early?
Should we allow the optimizer to add spurious
stores? #133

Open

rust-lang / unsafe-code-guidelines

Code Issues 214 Pull requests Actions Projects Security Insights

Storing an object as &Header, but reading the
data past the end of the header #256

New issue

Open

rust-lang / unsafe-code-guidelines

Code Issues 214 Pull requests Actions Projects Security Insights

Stacked Borrows: raw pointer usable only for T
too strict? #134

Open

New issue

rust-lang / unsafe-code-guidelines

Code Issues 214 Pull requests Actions Projects Security Insights

Stacked Borrows: Handling of two-phase
borrows #85

Closed



Confusion about stacked borrows

help



Inconsistency between stacked borrows and my mental
model

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Log In

Aug 2020

Tree Borrows

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Stacked Borrows incompatible with Garbage Collectors?

foconoco

Feb 2022

Feb 2022

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2022

Stacked Borrows cannot properly handle extern
type #276

Open

New issue

rust-lang / unsafe-code-guidelines

Code Issues 214 Pull requests Actions Projects Security Insights

Help me understand this UB with stacked borrows detected
by `miri`

help

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Apr 2022

1 / 14

Confusing stacked borrows violation #1364

Closed

New issue

rust-lang / miri

Code Issues 144 Pull requests 12 Actions Projects Security Insights

Anecdotal evidence supported by data:

- analysis of 30 000 libraries
- 6000+ tests have aliasing UB under Stacked Borrows

Tree Borrows allows much more code

Stacked Borrows (SB)

Tree Borrows uses a **tree** instead of a stack to track borrows

Out of 30 000 most downloaded libraries,
> **50% fewer tests** with aliasing UB when using Tree Borrows

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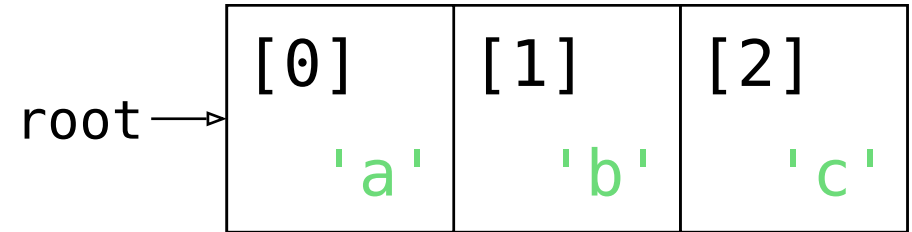
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fixes known technical limitations of SB,
incl. 2-phase borrows, extern types, **pointer offsets**

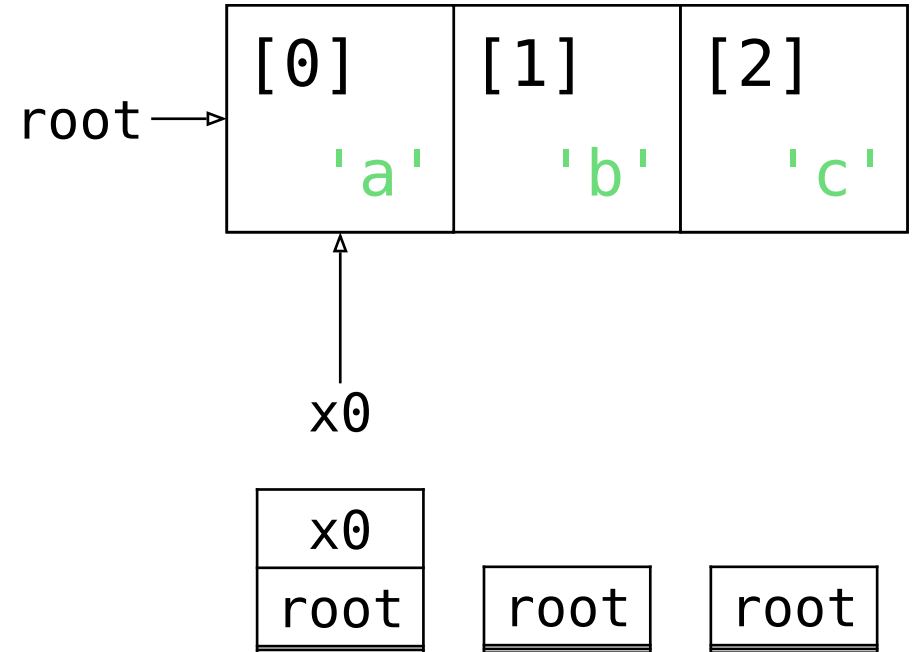
From Stacks to Trees

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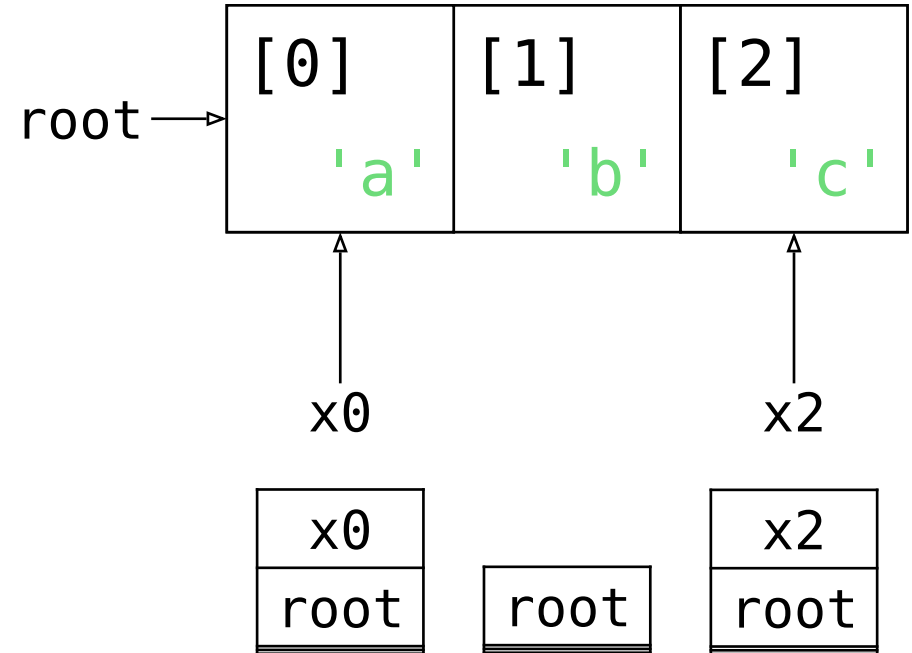
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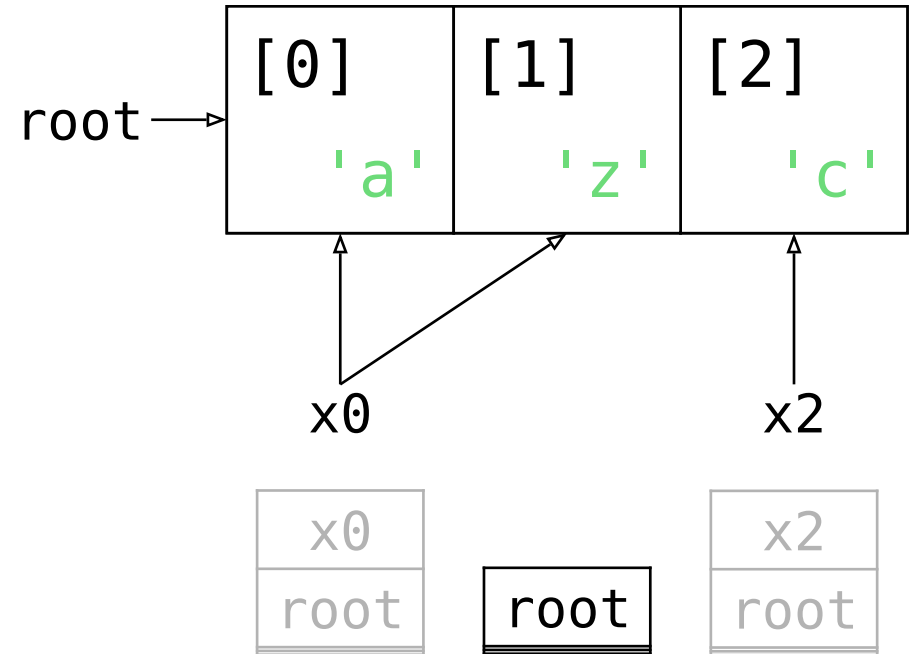
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// Scenario 1
unsafe { *x0.add(1) = 'z'; }

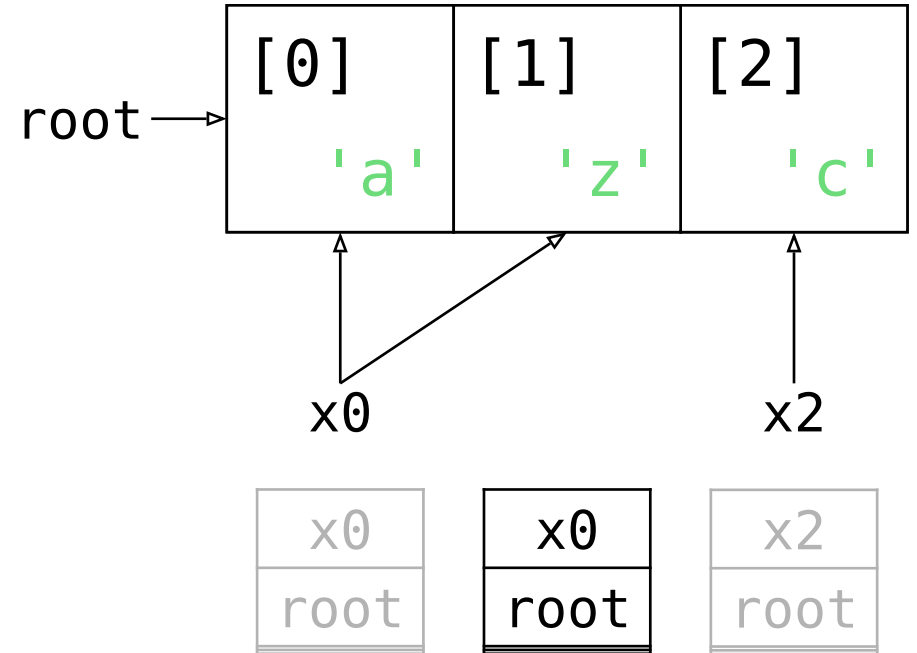
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Desired outcome: not UB



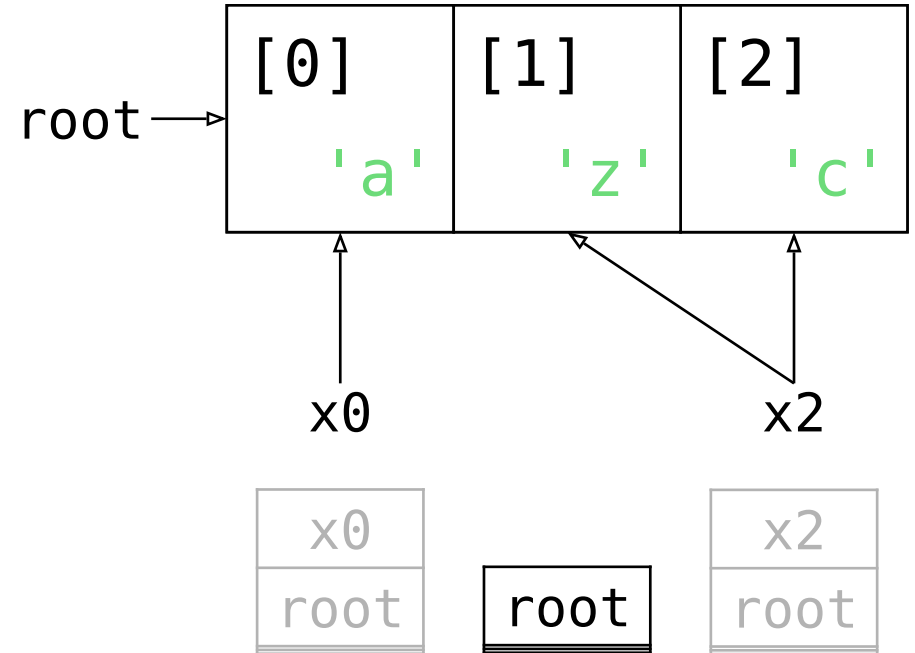

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// Scenario 2  
unsafe { *x2.sub(1) = 'z'; }
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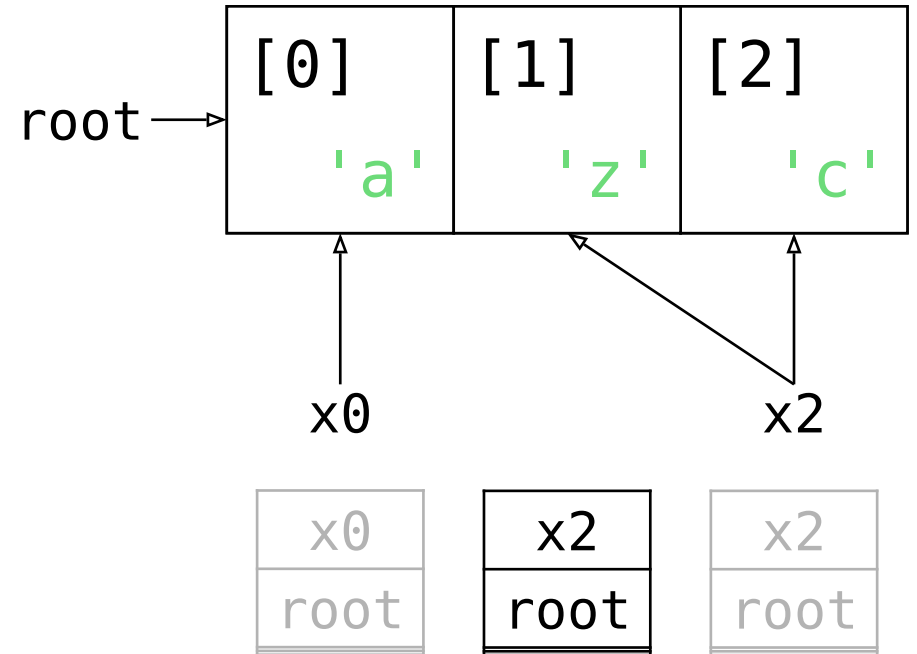


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```
// Scenario 2
```

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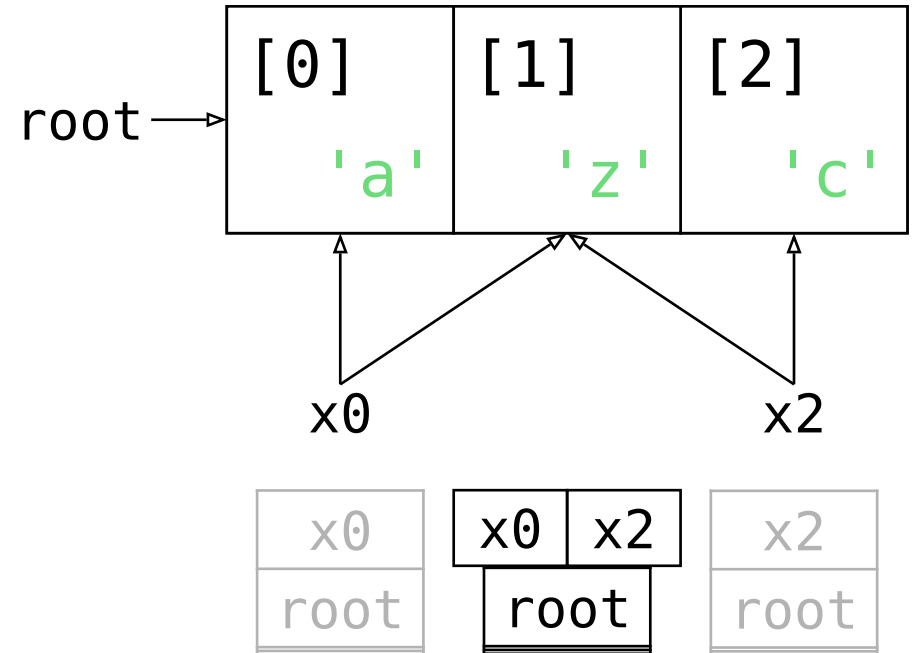
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let mut root = vec!['a', 'b', 'c'];
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// Scenario 1 or 2
unsafe { *??? = 'z'; }

```

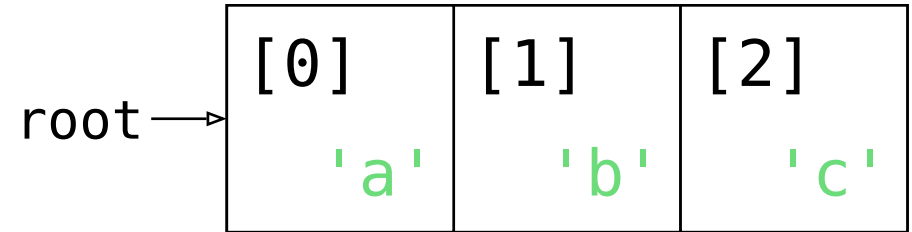
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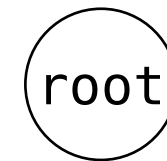
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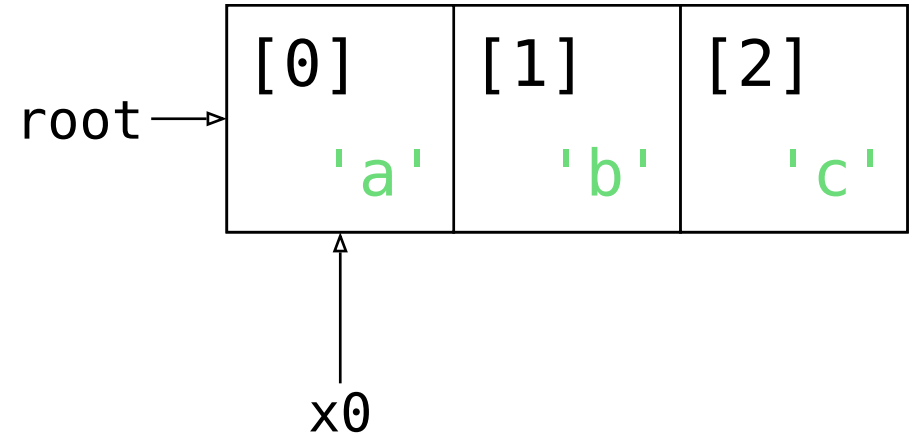
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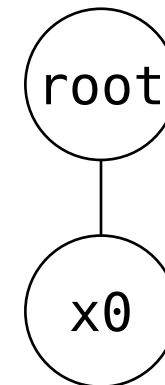
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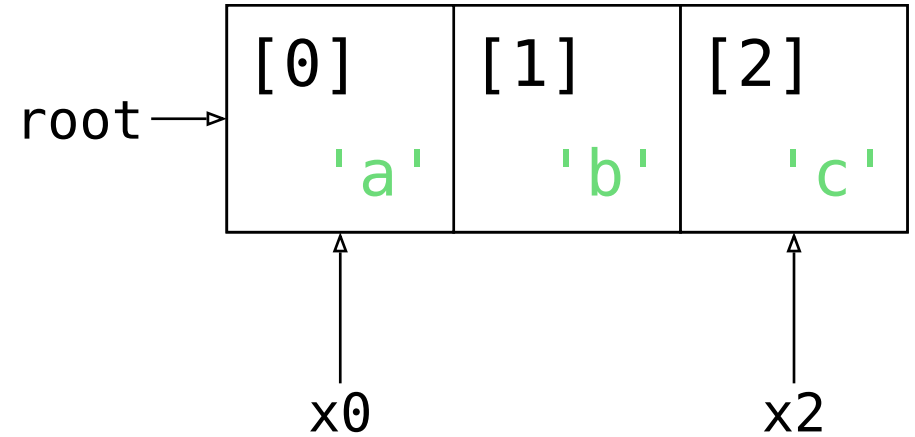
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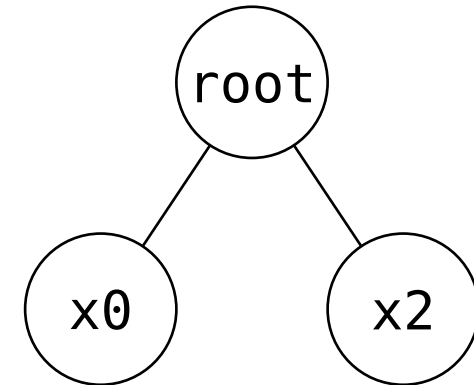
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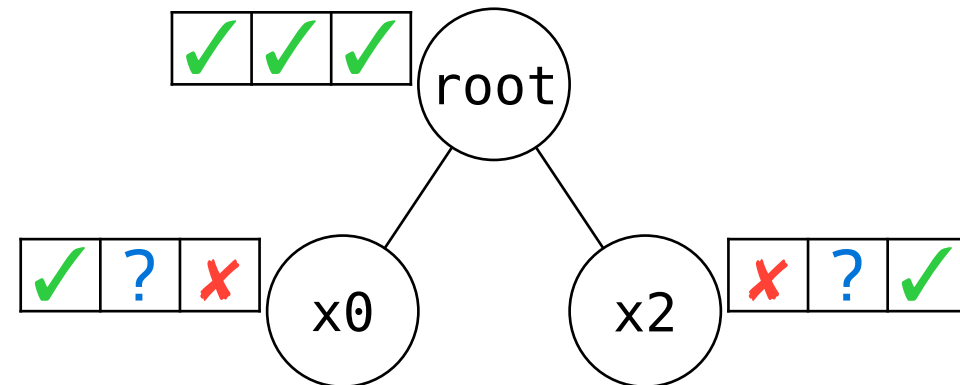
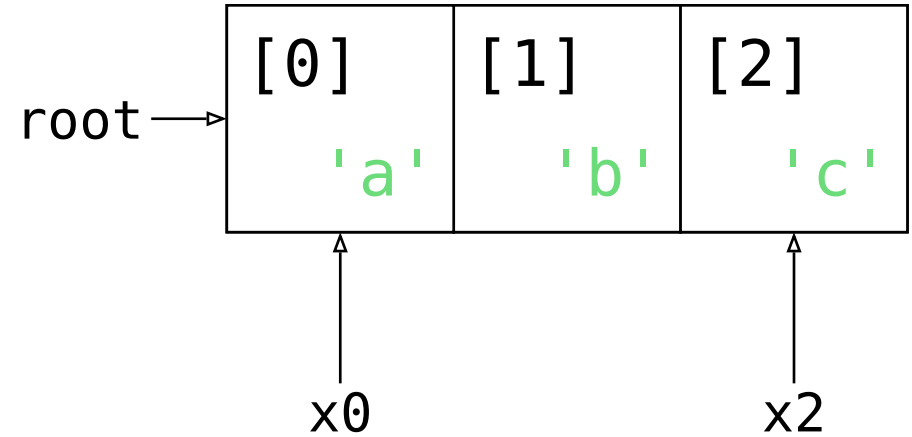
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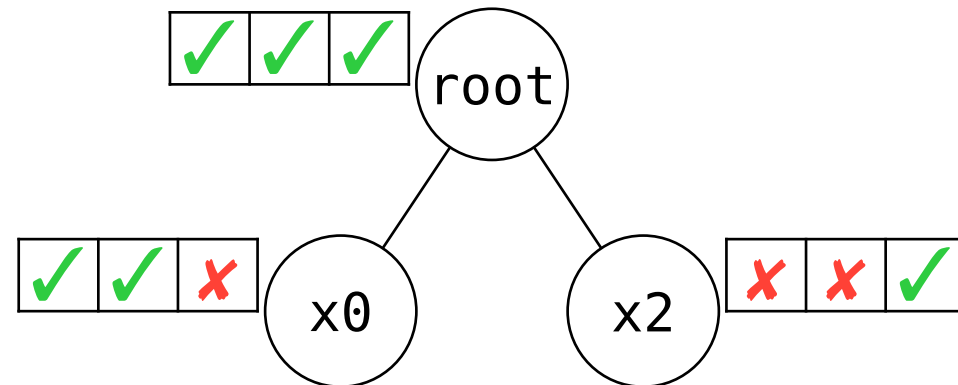
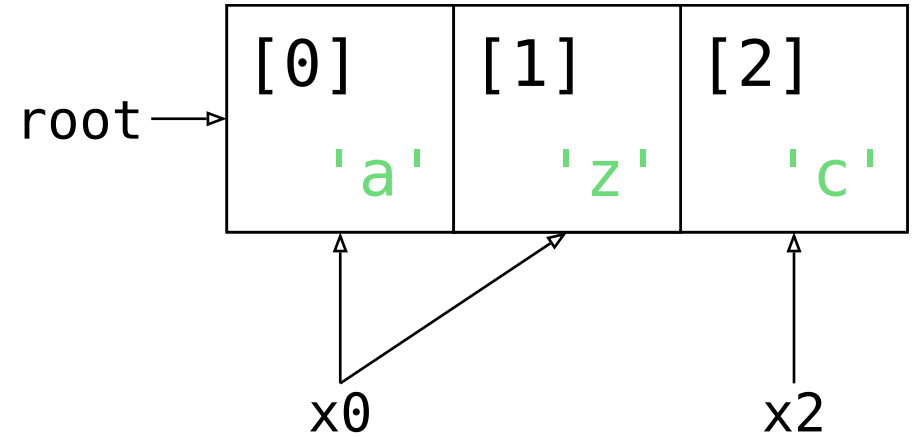
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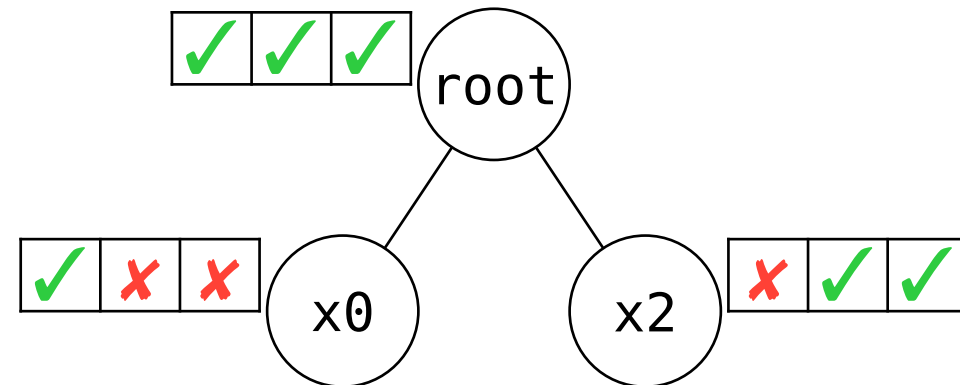
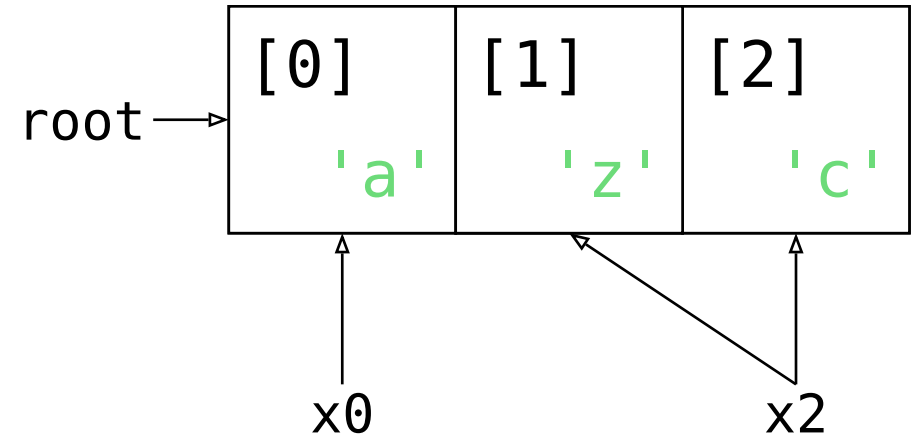


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let mut root = vec!['a', 'b', 'c'];
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// Scenario 2

```
unsafe { *x2.sub(1) = 'z'; }
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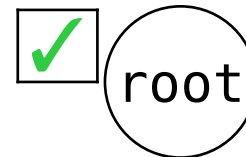
A second look at the motivating example

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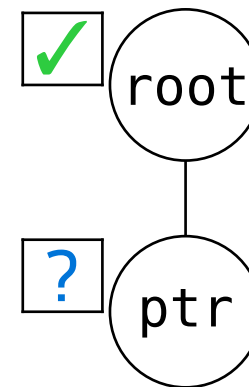
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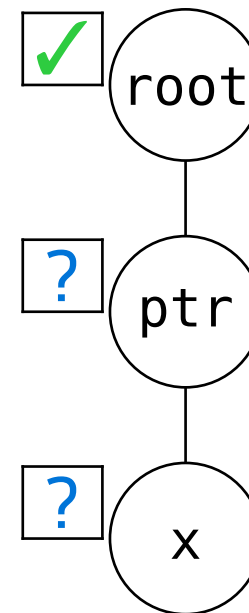


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From Stacks to Trees

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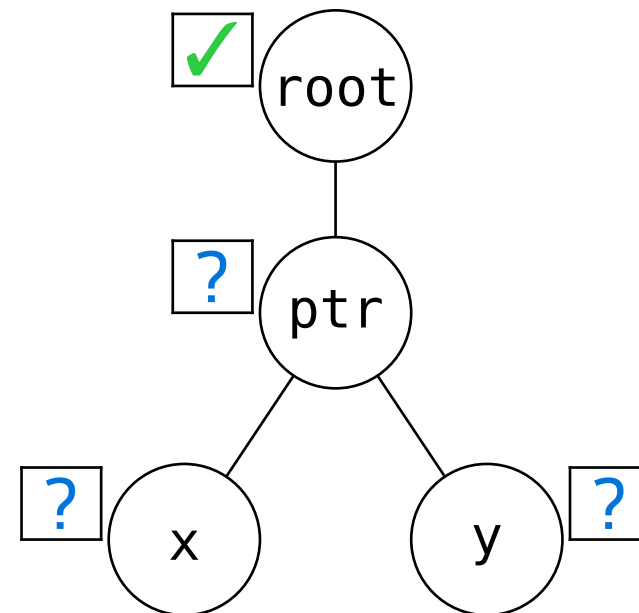


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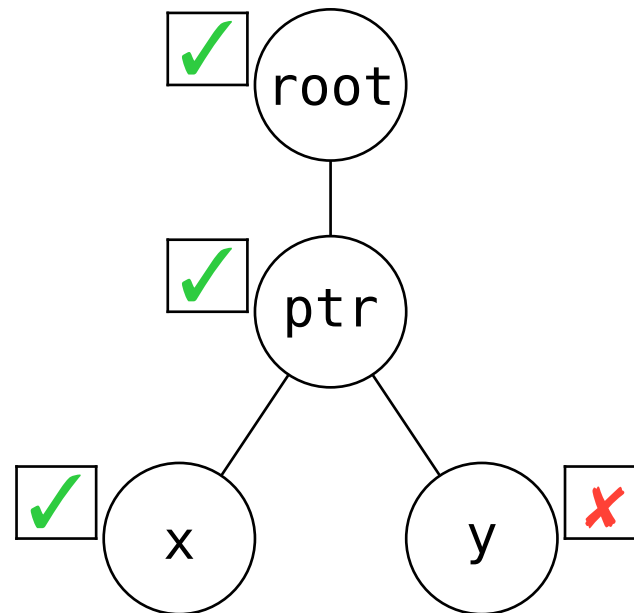
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let x = unsafe { &mut *ptr };  
let y = unsafe { &mut *ptr };  
// inline write_both(x, y):  
*x = 13;  
*y = 20;
```



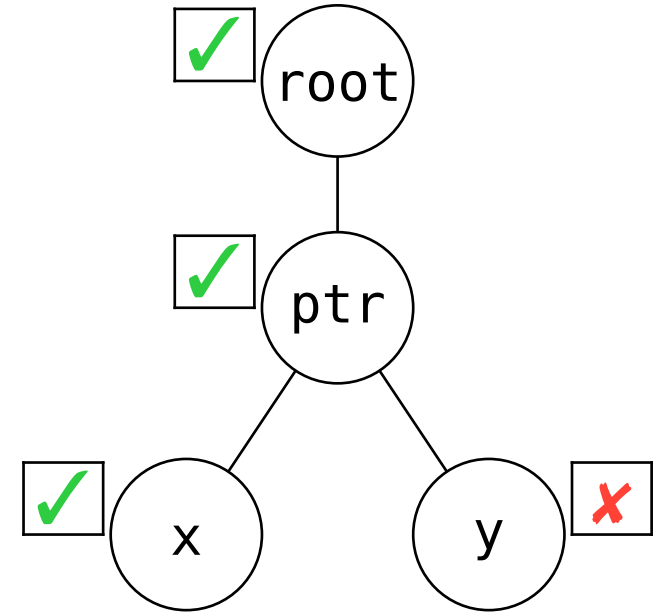
Desired outcome: UB

A second look at the motivating example

From Stacks to Trees

```
let mut root = 42;  
let ptr = &raw mut root;  
let x = unsafe { &mut *ptr };  
let y = unsafe { &mut *ptr };  
// inline write_both(x, y):  
*x = 13;  
*y = 20;
```

UB!



Desired outcome: UB

Evaluation

TB should enable desired optimizations

i.e. have enough UB to rule out problematic patterns

- formalized in Rocq (+Simuliris)
- a selection of optimizations proven
 - ✓ delete read through `&mut` or `&`
 - ✓ insert read through `&` in function
 - ✓ move read down for `&mut` or `&` in function
- ...

It should be possible to write **unsafe** code free of UB Evaluation

i.e. UB should be predictable and not too common

50% fewer tests have aliasing UB according to Tree Borrows

Only 31 ($< 0.01\%$) tests are regressions, all easily fixable.

(Out of 30 000 libraries, 400 000+ working tests)

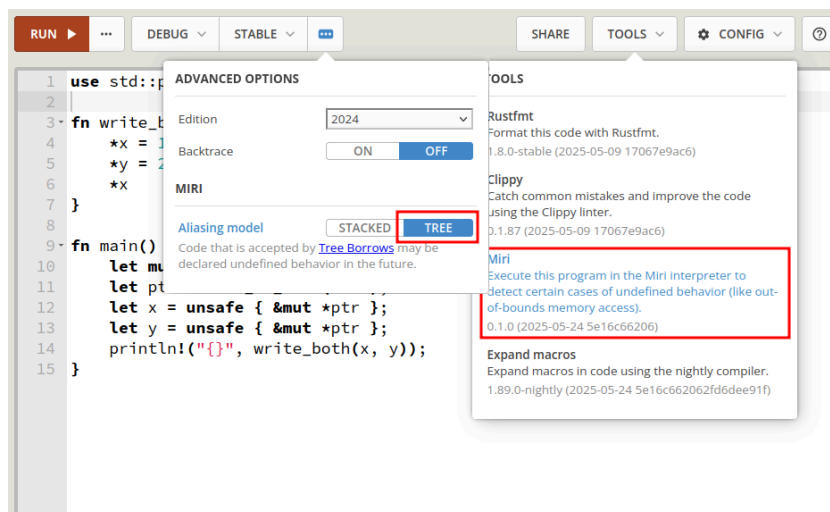
“Tree Borrows accepts more real-world programs that call foreign functions than Stacked Borrows due to differences in handling pointer arithmetic.”

A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries,
by I. McCormack, J. Sunshine, J. Aldrich @ ICSE'25

Conclusion

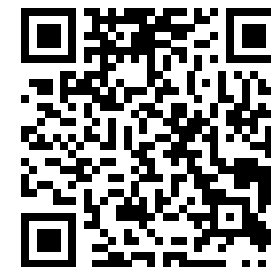
Try it out: available in Miri

Rust Playground supports TB
(play.rust-lang.org)



Learn more:

[plf.inf.ethz.ch/research/
pldi25-tree-borrows.html](https://plf.inf.ethz.ch/research/pldi25-tree-borrows.html)



Includes e.g. handling of raw pointers and interior mutability.

Postdoc positions available

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